

Getting Started With the R Commander*

John Fox

Version 2.8-0 (last modified: 2024-09-30)

1 Introduction

The **R Commander** (Fox, 2005, 2017) provides a graphical user interface (“GUI”) to the open-source **R** statistical computing environment (R Core Team, 2020). This manual is a brief, basic introduction to the **R Commander**; for more extensive documentation, see Fox (2017), which has a website at [<https://www.john-fox.ca/RCommander/>](https://www.john-fox.ca/RCommander/).

R is a command-driven system, and new users often find learning **R** challenging. This is particularly true of those who are new to statistical methods, such as students in basic-statistics courses. By providing a point-and-click interface to **R**, the **R Commander** allows these users to focus on statistical methods rather than on remembering and formulating **R** commands. Moreover, by rendering the generated commands visible to users, the **R Commander** has the potential for easing the transition to writing **R** commands, at least for some users. The **R Commander**, however, accesses only a small fraction of the capabilities of **R** and the literally thousands of **R** packages contributed by users to the Comprehensive R Archive Network (CRAN). The **R Commander** is itself extensible through plug-in packages, and many such plug-ins are now available on CRAN (see Section 6.4 of this document).

This document directly describes the use of the **R Commander** under the **Windows** version of **R**. There are small differences in the appearance and use of the **R Commander** under **macOS** and on **Linux** and **Unix** systems. Information about installing the **R Commander** on these platforms is available by following the link to the installation notes at the **R Commander** web page [<https://www.john-fox.ca/RCommander/>](https://www.john-fox.ca/RCommander/).

I use the following typographical conventions in this document: Names of software, such as **Windows**, **R**, the **Rcmdr** package, and the **R Commander**, are set in **boldface** type. The names of GUI elements such as menus, menu items, windows, and dialog boxes, are set in *italic* type. Variable names, names of data sets, and **R** commands are set in a **typewriter font**.

2 Starting the R Commander

Once **R** is running, simply loading the **Rcmdr** package by typing the command `library(Rcmdr)` into the *R Console* starts the **R Commander** graphical user interface. To function optimally under **Windows**, the **R Commander** prefers the single-document interface (“SDI”) to **R**.¹ After loading the package, *R Console* and *R Commander* windows should appear more or less as in Figures 1 and 2.² These and other screen

*Parts of this manual are adapted and updated from Fox (2005). Please address correspondence to current maintainer Manuel Munoz-Marquez manuel.munoz@uca.es.

¹The **Windows** version of **R** is normally run from a multiple-document interface (“MDI”), which contains the *R Console* window, *Graphical Device* windows created during the session, and other windows related to the **R** process. In contrast, under the single-document interface (“SDI”), the *R Console* and *Graphical Device* windows are not contained within a master window. There are several ways to run **R** in SDI mode — for example, by selecting the SDI when **R** is installed, by editing the `Rconsole` file in **R**’s `etc` subdirectory, or by adding `--sdi` to the *Target* field in the *Shortcut* tab of the **R** desktop icon’s *Properties*. You should be able to use the **R Commander** with the MDI, but it will not appear within the master **R** window and arranging the screen will be inconvenient.

²Most of the “screen-shots” in this document were produced with an earlier version of **R** and the **R Commander**. Screen images were only updated when their appearance or content has changed.

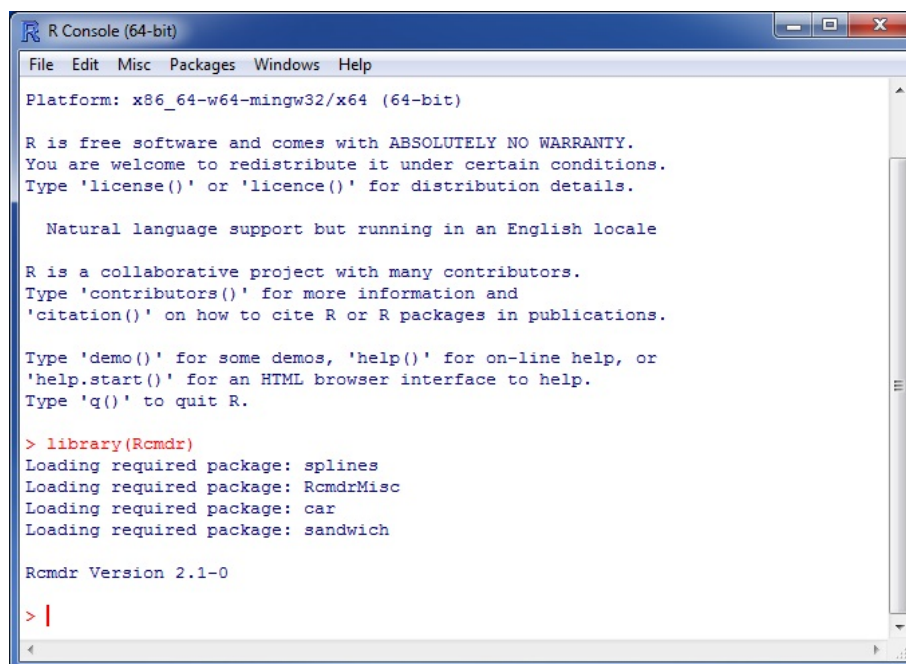


Figure 1: The *R Console* window after loading the **Rcmdr** package.

images in this document were created under **Windows 7** and **10**; if you use another version of **Windows** (or, of course, another computing platform), then the appearance of the screen may differ.³

The *R Commander* and *R Console* windows float freely on the desktop. You will normally use the **R Commander**’s menus and dialog boxes to read, manipulate, and analyze data, and you can safely minimize the *R Console* window.

- **R** commands generated by the **R Commander** GUI appear in the *R Script* tab in the upper pane of the main *R Commander* window. You can also type **R** commands directly into the script pane;⁴ the main purpose of the **R Commander**, however, is to avoid having to type commands. The second tab in the upper pane (labelled *R Markdown*) also accumulates the commands produced by the **R Commander** and can be used to generate printed reports; the *R Markdown* tab is described in Section 6.1.
- Printed output appears by default in the second pane (labelled *Output*).
- The lower, gray pane (labelled *Messages*) displays error messages, warnings, and some other information (“notes”), such as the start-up message in Figure 2.
- When you create graphs, these will appear in a separate *Graphics Device* window.

³Notice that the **R Commander** requires some packages in addition to several of the “recommended” packages that are normally distributed with **R**. The **Rcmdr** package, the required packages, and many other contributed packages are available for download from the Comprehensive R Archive Network (CRAN) at <<https://cran.r-project.org/>>.

If these packages are not installed, the **R Commander** will offer to install them from the internet or from local files (e.g., on a CD-ROM or USB flash drive). If you install the **Rcmdr** package via the **Windows** “R GUI,” not all of the packages on which the **Rcmdr** package depends will be installed. You can install the **Rcmdr** package and all of the packages on which it depends via the `install.packages` function, setting the argument `dependencies = TRUE`, but because of recursive dependencies, that will install more packages than are strictly necessary for the **R Commander** to function.

Thanks to Dirk Eddelbuettel, **Debian Linux** users need only issue the command `$ apt-get install r-cran-rcmdr` to install the **Rcmdr** package along with all of the packages that it requires. In any event, building and installing the **Rcmdr** package on **Linux** systems is typically straightforward. The task is a little more complicated under **macOS**, since the **tccltk** package on which the **Rcmdr** depends requires that **X-Windows** be installed — see the **R Commander** installation notes.

⁴You can also type commands at the `>` (greater-than) prompt in the *R Console*, but output generated by these commands will not appear in the **R Commander** *Output* pane and error and warning messages will normally not be visible.

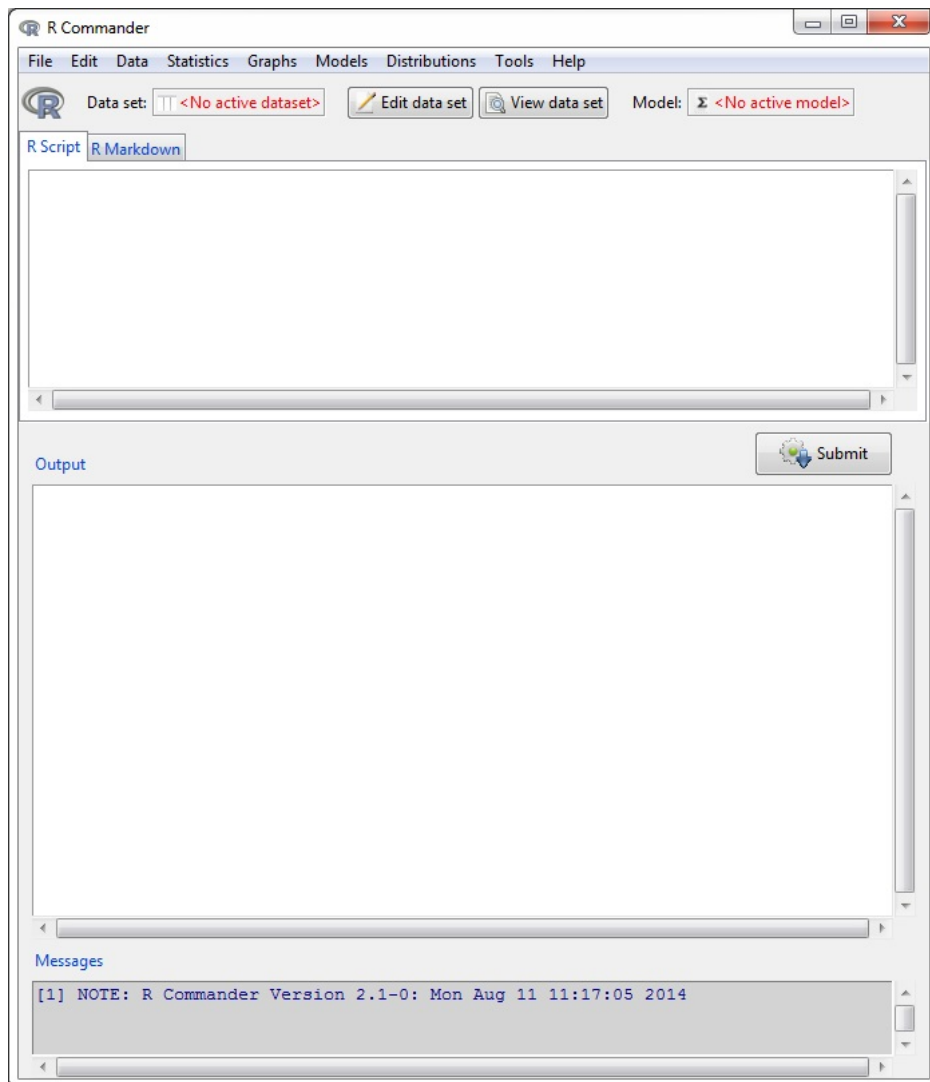


Figure 2: The *R Commander* window at start-up.

There are several menus along the top of the *R Commander* window:

File Menu items for loading and saving script files; for saving output and the **R** workspace; and for exiting.

Edit Menu items (*Cut, Copy, Paste*, etc.) for editing text in the various panes and tabs. Right-clicking in one of these panes or tabs also brings up an edit “context” menu.

Data Submenus containing menu items for reading and manipulating data.

Statistics Submenus containing menu items for a variety of statistical analyses.

Graphs Menu items for creating various statistical graphs.

Models Menu items and submenus for obtaining numerical summaries, confidence intervals, hypothesis tests, diagnostics, and graphs for a statistical model, and for adding diagnostic quantities, such as residuals, to the data set.

Distributions Submenus to obtain cumulative probabilities, probability densities or masses, quantiles, and graphs of standard statistical distributions (to be used, for example, as a substitute for statistical tables), and to generate samples from these distributions.

Tools Menu items for loading **R** packages unrelated to the **Rcmdr** package (e.g., to access data saved in another package); for loading **Rcmdr** plug-in packages (see Fox, 2017, Fox, 2007, Fox and Sá Carvalho, 2012, and Section 6.4 below); for setting most **R Commander** options and for saving options so that they will be applied in subsequent sessions; and for installing optional auxiliary software (see Section 6.1).

Help Menu items to obtain information about the **R Commander** (including this manual) and associated software. As well, each **R Commander** dialog box has a *Help* button (see below).

The complete menu “tree” for the **R Commander** (version 2.8-0) is shown below. Most menu items lead to dialog boxes, as illustrated later in this manual, and, as is conventional, such menu items are followed by Menu items are inactive (“grayed out”) if they are inapplicable to the current context. For example, if a data set contains no factors (categorical variables), the menu items for contingency tables will be inactive.⁵

File

- |– Change working directory...
- |– Open script file...
- |– Save script...
- |– Save script as...
- |– Open R Markdown file...
- |– Save R Markdown file...
- |– Save R Markdown file as...
- |– Save output...
- |– Save output as...
- |– Save R workspace...
- |– Save R workspace as...
- |– Exit
- | |– From Commander
- | |– From Commander and R
- | |– Restart the Commander

Edit

- |– Edit R Markdown document
- |– Edit knitr document

⁵Some menu items may not be displayed in certain circumstances. For example, the **knitr** menu items in the *File* menu will be displayed only if the *knitr* tab is activated, and the *Install auxiliary software* item in the *Tools* menu is activated only if the optional **L^AT_EX** or **Pandoc** software are missing. The menus also include dividers, which are not shown here.

- | - Remove last Markdown command block
- | - Remove last Markdown section title
- | - Remove last knitr command block
- | - Cut
- | - Copy
- | - Paste
- | - Delete
- | - Find
- | - Select all
- | - Undo
- | - Redo
- | - Clear window

Data

- | - New data set...
- | - Load data set...
- | - Merge data sets...
- | - Import data
 - | | - from text file, clipboard, or URL...
 - | | - from SPSS data set...
 - | | - from SAS xport file...
 - | | - from SAS b7dat file...
 - | | - from Minitab data set...
 - | | - from STATA data set...
 - | | - from Excel file...
- | - Data in packages
 - | | - List data sets in packages
 - | | - Read data set from an attached package...
- | - Active data set
 - | | - View data...
 - | | - Select active data set...
 - | | - Refresh active data set
 - | | - Help on active data set (if available)
 - | | - Variables in active data set
 - | | - Set case names...
 - | | - Subset active data set...
 - | | - Sort active data set...
 - | | - Aggregate variables in active data set...
 - | | - Select row(s) from active data set...
 - | | - Remove row(s) from active data set...
 - | | - Stack variables in active data set...
 - | | - Remove cases with missing data...
 - | | - Reshape data set from long to wide format...
 - | | - Reshape data set from wide to long format...
 - | | - Convert all character variables to factors...
 - | | - Save active data set...
 - | | - Export active data set...
- | - Manage variables in active data set
 - | | - Recode variables...
 - | | - Compute new variable...
 - | | - Add observation numbers to data set
 - | | - Standardize variables...
 - | | - Convert numeric variables to factors...
 - | | - Convert character variables to factors...
 - | | - Bin numeric variable...

- | |- Reorder factor levels...
- | |- Drop unused factor levels...
- | |- Define contrasts for a factor...
- | |- Rename variables...
- | |- Delete variables from data set ...

Statistics

- | - Summaries
 - | |- Active data set
 - | |- Numerical summaries...
 - | |- Frequency distributions...
 - | |- Count missing observations
 - | |- Table of statistics...
 - | |- Correlation matrix...
 - | |- Correlation test...
 - | |- Test of normality...
 - | |- Transform toward normality...
- | - Contingency tables
 - | |- Two-way table...
 - | |- Multi-way table...
 - | |- Enter and analyze two-way table...
- | - Means
 - | |- Single-sample t-test...
 - | |- Independent samples t-test...
 - | |- Paired t-test...
 - | |- One-way ANOVA...
 - | |- Multi-way ANOVA...
 - | |- One-factor repeated-measures ANOVA/ANCOVA...
 - | |- Two-factor repeated-measures ANOVA/ANCOVA...
- | - Proportions
 - | |- Single-sample proportion test...
 - | |- Two-sample proportions test...
- | - Variances
 - | |- Two-variances F-test...
 - | |- Bartlett's test...
 - | |- Levene's test...
- | - Nonparametric tests
 - | |- Two-sample Wilcoxon test...
 - | |- Single-sample Wilcoxon test...
 - | |- Paired-samples Wilcoxon test...
 - | |- Kruskal-Wallis test...
 - | |- Friedman rank-sum test...
- | - Dimensional analysis
 - | |- Scale reliability...
 - | |- Principal-components analysis...
 - | |- Factor analysis...
 - | |- Confirmatory factor analysis...
 - | |- Cluster analysis
 - | | |- k-means cluster analysis...
 - | | |- Hierarchical cluster analysis...
 - | | |- Summarize hierarchical clustering...
 - | | |- Add hierarchical clustering to data set...
- | - Fit models
 - | |- Linear regression...
 - | |- Linear model...

- | |- Generalized linear model...
- | |- Multinomial logit model...
- | |- Ordinal regression model...
- | |- Linear mixed model...
- | |- Generalized linear mixed model...

Graphs

- | - Color palette...
- | - Index plot...
- | - Dot plot...
- | - Histogram...
- | - Plot discrete numeric variable.....
- | - Density estimate...
- | - Stem-and-leaf display...
- | - Boxplot...
- | - Quantile-comparison plot...
- | - Symmetry boxplot...
- | - Scatterplot...
- | - Scatterplot matrix...
- | - Line graph...
- | - XY conditioning plot...
- | - Plot of means...
- | - Strip chart...
- | - Bar graph...
- | - Pie chart...
- | - 3D graph
 - | |- 3D scatterplot...
 - | |- Identify observations with mouse
 - | |- Save graph to file
- | - Save graph to file
 - | |- as bitmap...
 - | |- as PDF/Postscript/EPS...
- | |- 3D RGL graph...

Models

- | - Select active model...
- | - Summarize model
- | - Compare model coefficients...
- | - Add observation statistics to data...
- | - Akaike Information Criterion (AIC)
- | - Bayesian Information Criterion (BIC)
- | - Stepwise model selection...
- | - Subset model selection...
- | - Confidence intervals.....
- | - Bootstrap confidence intervals...
- | - Delta method confidence interval...
- | - Hypothesis tests
 - | |- ANOVA table...
 - | |- Compare two models...
 - | |- Linear hypothesis...
- | - Numerical diagnostics
 - | |- Variance-inflation factors
 - | |- Breusch-Pagan test for heteroscedasticity...
 - | |- Durbin-Watson test for autocorrelation...
 - | |- RESET test for nonlinearity...
- | |- Bonferroni outlier test

- | |- Response transformation
- | - Graphs
 - | |- Basic diagnostic plots
 - | |- Residual quantile-comparison plot...
 - | |- Component+residual plots...
 - | |- 3D component+residual plot...
 - | |- Added-variable plots...
 - | |- 3D added-variable plot...
 - | |- Influence plot...
 - | |- Influence index plot...
 - | |- Effect plots...
 - | |- Predictor effect plots...
- Distributions
 - | - Set random number generator seed...
 - | - Continuous distributions
 - | |- Normal distribution
 - | | |- Normal quantiles...
 - | | |- Normal probabilities...
 - | | |- Plot normal distribution...
 - | | |- Sample from normal distribution...
 - | |- t distribution
 - | | |- t quantiles...
 - | | |- t probabilities...
 - | | |- Plot t distribution...
 - | | |- Sample from t distribution...
 - | |- Chi-squared distribution
 - | | |- Chi-squared quantiles...
 - | | |- Chi-squared probabilities...
 - | | |- Plot chi-squared distribution...
 - | | |- Sample from chi-squared distribution...
 - | |- F distribution
 - | | |- F quantiles...
 - | | |- F probabilities...
 - | | |- Plot F distribution...
 - | | |- Sample from F distribution...
 - | |- Exponential distribution
 - | | |- Exponential quantiles...
 - | | |- Exponential probabilities...
 - | | |- Plot exponential distribution...
 - | | |- Sample from exponential distribution...
 - | |- Uniform distribution
 - | | |- Uniform quantiles...
 - | | |- Uniform probabilities...
 - | | |- Plot uniform distribution...
 - | | |- Sample from uniform distribution...
 - | |- Beta distribution
 - | | |- Beta quantiles...
 - | | |- Beta probabilities...
 - | | |- Plot beta distribution...
 - | | |- Sample from beta distribution...
 - | |- Cauchy distribution
 - | | |- Cauchy quantiles...
 - | | |- Cauchy probabilities...
 - | | |- Plot Cauchy distribution...

```

| | |- Sample from Cauchy distribution...
| | |- Logistic distribution
| | |- Logistic quantiles...
| | |- Logistic probabilities...
| | |- Plot logistic distribution...
| | |- Sample from logistic distribution...
| | |- Lognormal distribution
| | |- Lognormal quantiles...
| | |- Lognormal probabilities...
| | |- Plot lognormal distribution...
| | |- Sample from lognormal distribution...
| | |- Gamma distribution
| | |- Gamma quantiles...
| | |- Gamma probabilities...
| | |- Plot gamma distribution...
| | |- Sample from gamma distribution...
| | |- Weibull distribution
| | |- Weibull quantiles...
| | |- Weibull probabilities...
| | |- Plot Weibull distribution...
| | |- Sample from Weibull distribution...
| | |- Gumbel distribution
| | |- Gumbel quantiles...
| | |- Gumbel probabilities...
| | |- Plot Gumbel distribution...
| | |- Sample from Gumbel distribution...
| | |- Discrete distributions
| | |- Binomial distribution
| | |- Binomial quantiles...
| | |- Binomial tail probabilities...
| | |- Binomial probabilities...
| | |- Plot binomial distribution...
| | |- Sample from binomial distribution...
| | |- Poisson distribution
| | |- Poisson quantiles...
| | |- Poisson tail probabilities...
| | |- Poisson probabilities...
| | |- Plot Poisson distribution...
| | |- Sample from Poisson distribution...
| | |- Geometric distribution
| | |- Geometric quantiles...
| | |- Geometric tail probabilities...
| | |- Geometric probabilities...
| | |- Plot geometric distribution...
| | |- Sample from geometric distribution...
| | |- Hypergeometric distribution
| | |- Hypergeometric quantiles...
| | |- Hypergeometric tail probabilities...
| | |- Hypergeometric probabilities...
| | |- Plot hypergeometric distribution...
| | |- Sample from hypergeometric distribution...
| | |- Negative binomial distribution
| | |- Negative binomial quantiles...
| | |- Negative binomial tail probabilities...

```

```

| | |- Negative binomial probabilities...
| | |- Plot negative binomial distribution...
| | |- Sample from negative binomial distribution...
Tools
|- Load package(s)...
|- Load Rcmdr plug-in(s)...
|- Options...
|- Save Rcmdr options...
|- Manage Mac OS X app nap for R.app...
|- Install auxiliary software...
Help
|- Commander help
|- Introduction to the R Commander
|- R Commander website
|- About Rcmdr
|- R Commander hex sticker
|- Help on active data set (if available)
|- Start R help system
|- R website
|- Using R Markdown

```

The **R Commander** interface includes a few elements in addition to menus and dialogs:

- Below the menus is a “toolbar” with a row of buttons.
 - The left-most (flat) button shows the name of the active data set. Initially there is no active data set. If you press this button, you will be able to choose among data sets currently in memory (if there is more than one). Most of the menus and dialogs in the **R Commander** reference the active data set. (The *File*, *Edit*, and *Distributions* menus are exceptions.)
 - Two buttons allow you to open the **R Commander** data editor to modify the active data set or a viewer to examine it.⁶ The data-set viewer can remain open while other operations are performed, and the view is refreshed if the data set is modified, for example by adding a variable.⁷
 - A flat button indicates the name of the active statistical model — a linear model (such as a linear-regression model), a generalized linear model, a multinomial logit model, an ordinal regression model, or a linear or generalized linear mixed-effects model.⁸ Initially there is no active model. If there is more than one model in memory associated with the active data set, you can choose among the models by pressing the button. The **R Commander** synchronizes models and the data sets to which they are fit.
- Immediately below the toolbar is a pane containing the *R Script* tab, a large scrollable text window. As mentioned, commands generated by the **R Commander** are copied into this window. You can edit the text in the *Script* tab or even type your own **R** commands into the window. Pressing the *Submit* button, which is at the right below the *Script* tab (or, alternatively, the key combination *Ctrl-r*,⁹ for “run,” or *Ctrl-Tab*), causes the line containing the cursor to be submitted (or resubmitted) for execution. If several lines are selected (e.g., by left-clicking and dragging the mouse over them), then pressing *Submit* will cause all of them to be executed. Commands entered into the *R Script* tab can

⁶By default, the **R Commander** data editor is used if the number of values (cells) in the data set is 10,000 or fewer; for larger data sets, the standard **R** editor is used. See the **R Commander** help for information on resetting this threshold.

⁷The data viewer, provided by the `showData` function from David Firth’s **relimp** package (Firth, 2016), can be slow for data sets with large numbers of variables. When the number of variables exceeds 100 or the number of cases exceeds 20,000, the less aesthetically pleasing **R View** command is used instead to display the data set. To use **View** regardless of the number of variables or cases, set these thresholds to 0. See the **R Commander** help for details.

⁸**R Commander** plug-in packages (Fox, 2017; Fox, 2007; Fox and Sá Carvalho, 2012) may provide additional classes of models.

⁹That is, hold down the *Ctrl* (or *Control*) key and simultaneously press the *r* key. On **macOS** you can usually substitute the *Command* key for the *Control* key, both here and more generally, but *Control* key combinations can be used as well.

extend over more than one line, but all lines must be submitted simultaneously. The key combination *Ctrl-a* selects all of the text in the *Script* tab, and *Ctrl-s* brings up a dialog box to save the contents of the tab. The *R Markdown* tab is described in Section 6.1.

- Below the *R Script* and *R Markdown* tabs is a pane containing a large scrollable and editable text window for *Output*. Commands echoed to the *Output* pane appear in red, the resulting output in dark blue (as in the standard **Windows R Console**).
- At the bottom is a small gray pane for *Messages*. Error messages are displayed in red text, warnings in green, and other messages in dark blue. Errors and warnings also provide an audible cue by ringing a bell.

As mentioned, once you have loaded the **Rcmdr** package, you can minimize the *R Console*. The *R Commander* window can also be resized or maximized in the normal manner. If you resize the **R Commander**, the width of subsequent **R** output is automatically adjusted to fit the *Output* pane.

The **R Commander** is highly configurable: I have described the default configuration here. Changes to the configuration can be made via the *Tools* → *Options...* menu, or — more extensively — by setting **R Commander** options in **R**.¹⁰ See *Help* → *Commander help* for details.

3 Data Input

Most of the procedures in the **R Commander** assume that there is an active data set.¹¹ If there are several data sets in memory, you can choose among them, but only one is active. When the **R Commander** starts up, there is no active data set.

The **R Commander** provides several ways to get data into **R**:

- Using the **R Commander** data editor, You can enter data directly via *Data* → *New data set...* This is a reasonable choice only for a very small data set.
- You can import data from a plain-text (“ascii”) file or the clipboard, over the internet from a URL, from another statistical package (**Minitab**, **SPSS**, **SAS**, or **Stata**), or from an **Excel** spreadsheet.
- You can read a data set that is included in an **R** package, either typing the name of the data set (if you know it), or selecting the data set in a dialog box.

3.1 Reading Data From a Text File

For example, consider the data file *Nations.txt*.¹² The first few lines of the file are as follows:

```
TFR contraception infant.mortality GDP region
Afghanistan      6.90   NA  154  2848   Asia
Albania          2.60   NA   32   863   Europe
Algeria          3.81   52   44  1531   Africa
American-Samoa   NA     NA   11   NA     Oceania
Andorra          NA     NA   NA   NA     Europe
Angola           6.69   NA  124   355   Africa
Antigua          NA     53   24  6966   Americas
Argentina        2.62   NA   22  8055   Americas
Armenia          1.70   22   25   354   Europe
Australia        1.89   76    6 20046   Oceania
. . .
```

¹⁰A menu item that terminates in ellipses (i.e., three dots, ...) leads to a dialog box; this is a standard GUI convention. In this document, → represents selecting a menu item or submenu from a menu.

¹¹Procedures selected under via the *Distributions* menu are exceptions, as is *Enter and analyze two-way table...* under the *Statistics* → *Contingency tables* menu.

¹²This file resides in the *etc* subdirectory of the **Rcmdr** package. The data are for 1998 and are from the United Nations.

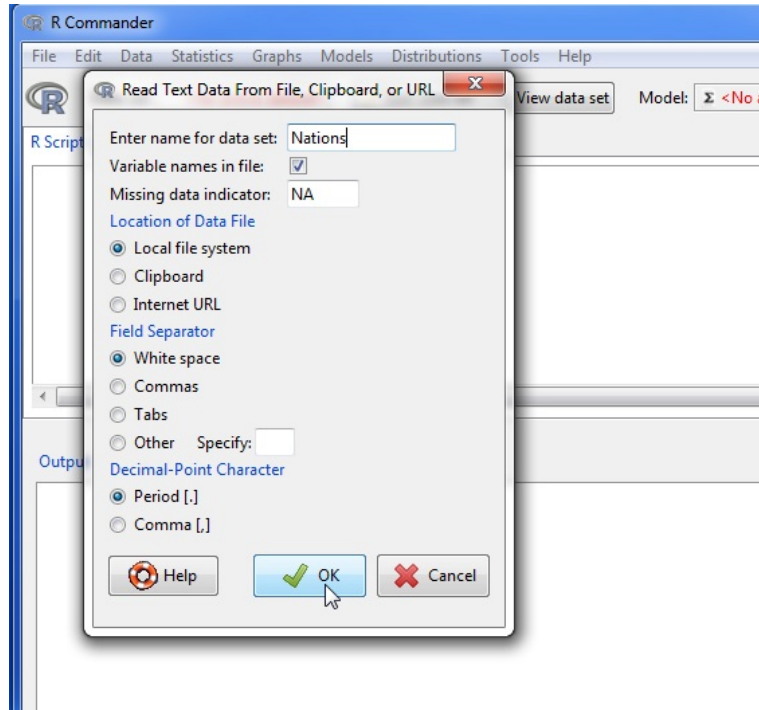


Figure 3: Reading data from a text file.

- The first line of the file contains variable names: **TFR** (the total fertility rate, expressed as number of children per woman), **contraception** (the rate of contraceptive use among married women, in percent), **infant.mortality** (the infant-mortality rate per 1000 live births), **GDP** (gross domestic product per capita, in U.S. dollars), and **region**.
- Subsequent lines contain the data values themselves, one line per country. The data values are separated by “white space” — one or more blanks or tabs. Although it is helpful to make the data values line up vertically, it is not necessary to do so. Notice that the data lines begin with the country names. Because I want these to be the “row names” for the data set, there is no corresponding variable name: That is, there are five variable names but six data values on each line, the first of which is alphabetic. When this happens, the **R** `read.table` command will interpret the first value on each line as the row name.
- Some of the data values are missing. In **R**, it is most convenient to use **NA** (representing “not available”) to encode missing data, as I have done here.
- The variables **TFR**, **contraception**, **infant.mortality**, and **GDP** are numeric (quantitative) variables; in contrast, **region** contains region names. When the data are read, **R** will treat **region** as a “factor” — that is, as a categorical variable. In most contexts, the **R Commander** distinguishes between numerical variables and factors, and will try to prevent you from doing unreasonable things, such as computing the mean of a factor.¹³

To read the **Nations.txt** data file into **R**, select *Data* → *Import data* → *from text file, clipboard, or URL...* from the *R Commander* menus. This operation brings up a *Read Text Data* dialog, as shown in Figure 3. The default name of the data set is **Dataset**. I have changed the name to **Nations**.

Valid **R** names begin with an upper- or lower-case letter (or a period, `.`) and consist entirely of letters, periods, underscores (`_`), and numerals (i.e., 0–9); in particular, do not include any embedded blanks in a

¹³Character variables, whose values are character strings, and logical variables, whose values are **TRUE** or **FALSE** are treated as factors by the **R Commander**.

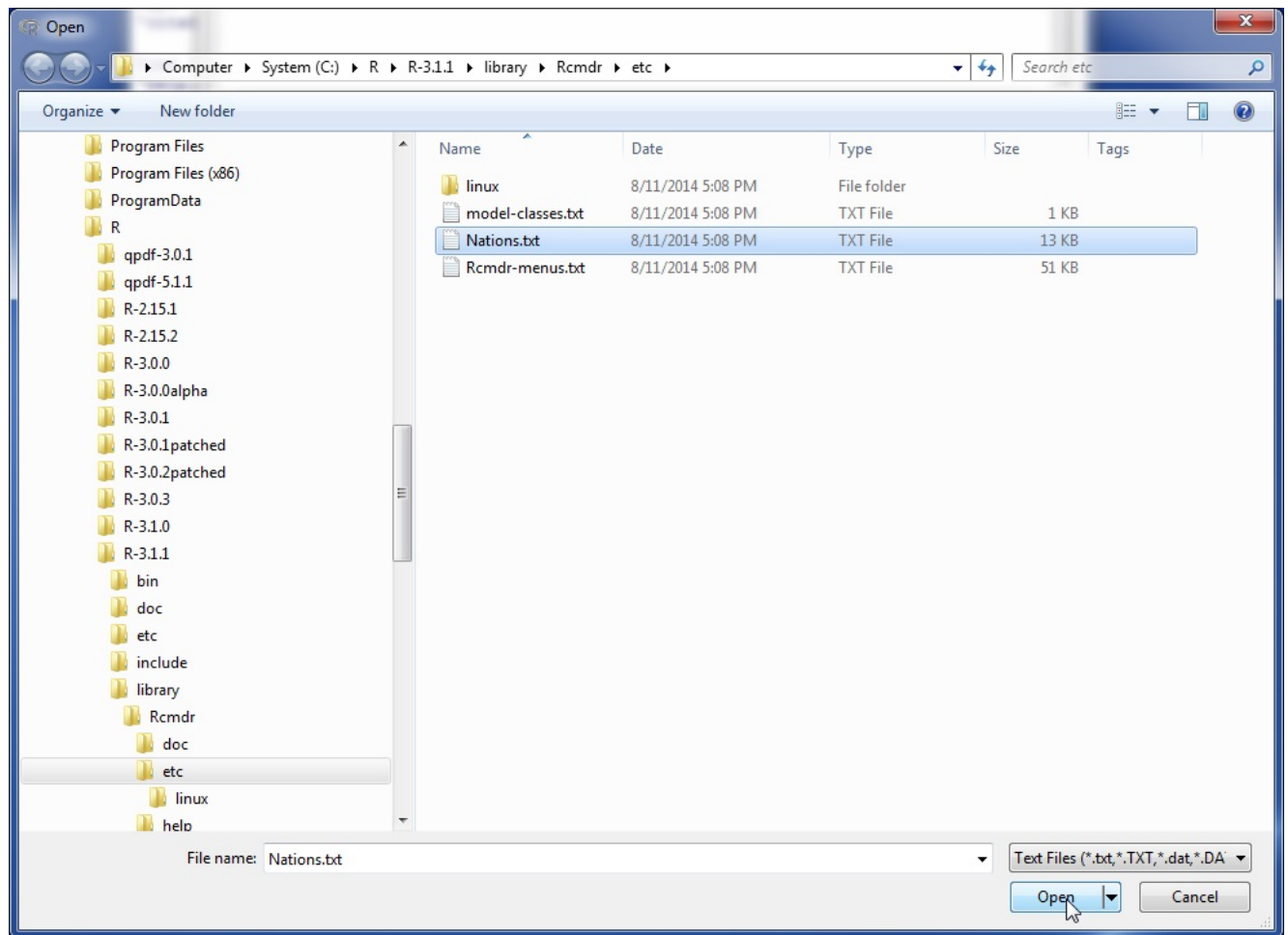


Figure 4: Open-file dialog for reading a text data file.

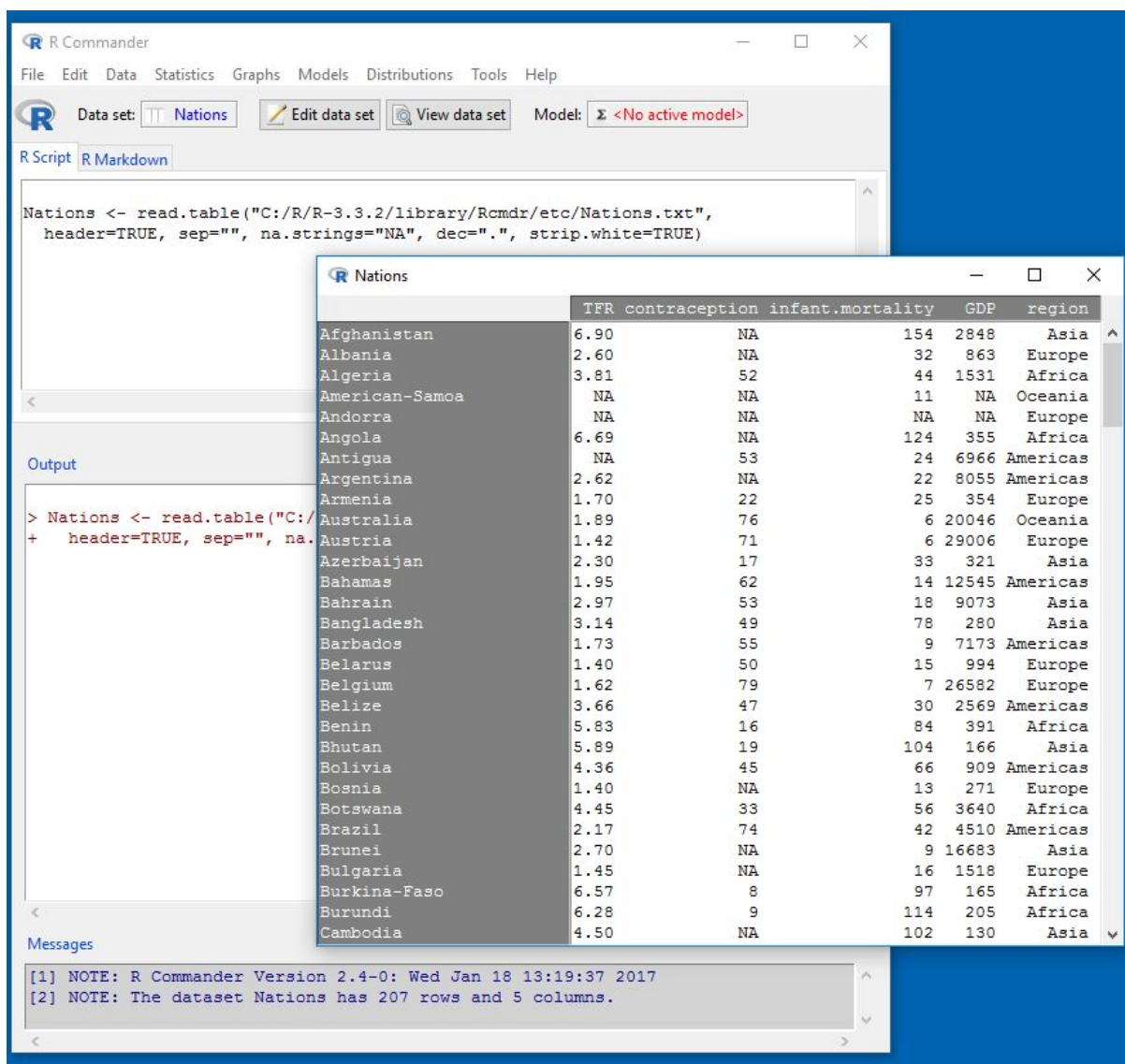


Figure 5: Displaying the active data set.

data-set name. **R** is case-sensitive, and so, for example, `nations`, `Nations`, and `NATIONS` are distinguished, and could be used to represent different data sets.

Clicking the *OK* button in the *Read Text Data* dialog brings up an *Open* file dialog, shown in Figure 4. Here I navigated to and selected the file `Nations.txt`. Clicking the *Open* button in the dialog causes the data file to be read. Once the data file is read, it becomes the active data set in the **R Commander**. As a consequence, in Figure 5, the name of the data set appears in the data set button near the top left of the *R Commander* window. The command to read the `Nations` data set (the **R** `read.table` command) appears in the *R Script* tab and *Output* pane. As well, when the data set is read and becomes the active data set, a note appears in the *Messages* pane.

I next clicked the *View data set* button to bring up the data viewer window, also shown in Figure 5.

The `read.table` command creates an **R** “data frame,” which is an object containing a rectangular cases-by-variables data set: The rows of the data set represent cases or observations and the columns represent variables. Data sets in the **R Commander** are **R** data frames.

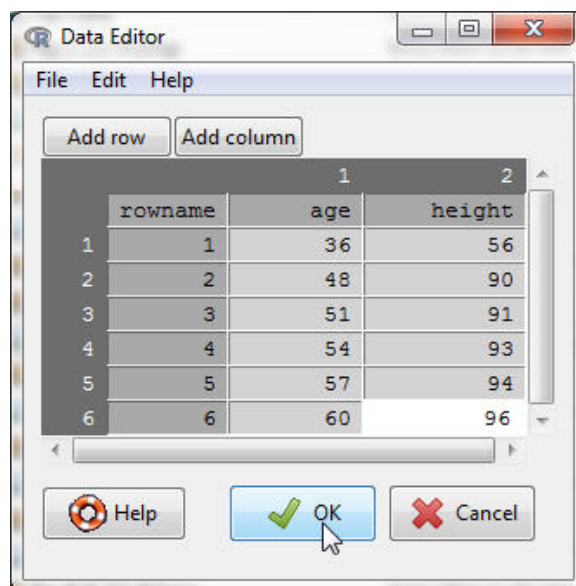


Figure 6: Data editor after the data are entered.

3.2 Entering Data Directly

You can enter data directly into the **R Commander** basic spreadsheet-like data editor. A simple alternative, which I in fact prefer, is to save the data in a plain-text file (be careful, if you create the data file with a word processor, to save it as a plain-text or “ascii” file), typically with file type `.txt`, and then to read the file as in the preceding section, via *Data* → *Import data* → *from text file, clipboard, or URL...*. If your data are already in a spreadsheet program, such as **Excel**, you can simply export the data to a comma-separated-values text file (`.csv` file), and read the file into the R Commander, being careful to specify the field separator as a comma. Recall that you can also read an **Excel** spreadsheet directly.

As an example of direct data input, I use a very small data set from Problem 2.44 in Moore (2000):

- Select *Data* → *New data set...* from the *R Commander* menus. Optionally enter a name for the data set, such as **Problem2.44**, in the resulting dialog box, and click the *OK* button. (Remember that **R** names cannot include embedded blanks.) This will bring up a *Data Editor* window with an empty data set.
- Enter the data from the problem into the first two columns of the data editor. Add a column by clicking the *Add column* button in the data editor toolbar, or by selecting *Add column* from the *Edit* menu. Similarly, add rows to the data set by clicking the *Add row* button repeatedly or via the *Edit* menu. You can also add a row by pressing the *Enter* key or add a column by pressing the *Tab* key when the cursor is in a cell in the data table.
- You can move from one cell to another by using the arrow keys on your keyboard or by pointing with the mouse and left-clicking. Originally, the variables are named `var1` and `var2`, and the data values are all `NA` (i.e., missing). When you type a new variable name, row name, or data value into a cell of the data editor, the new value replaces what was previously there. If you double-click in a cell, then the cell becomes `NA`. When you are finished entering the data, the data-editor window should look like Figure 6.
- In this example, both variables are numeric. If you type any non-numeric values in a column in the data editor (other than the missing value `NA`), then the column will define a factor (categorical variable) in the new data set.
- Select *File* → *Exit and save* from the *Data Editor* menus or click the *OK* button. The data set that you entered is now the active data set in the **R Commander**.

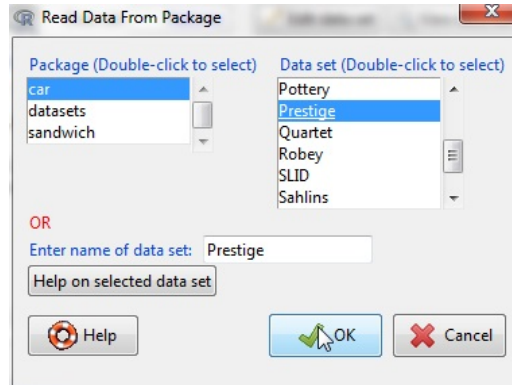


Figure 7: Reading data from an attached package — in this case the `Prestige` data set from the `car` package.

3.3 Reading Data from a Package

Many **R** packages include data. Data sets in packages can be listed in a pop-up window via *Data* → *Data in packages* → *List data sets in packages*, and can be read into the **R Commander** via *Data* → *Data in packages* → *Read data set from an attached package*.¹⁴ The resulting dialog box is shown in Figure 7.¹⁵ If you know the name of a data set in a package then you can enter its name directly; otherwise double-clicking on the name of a package displays its data sets in the right list box; and double-clicking on a data set name copies the name to the data-set entry field in the dialog.¹⁶ Pressing a letter key in the *Data set* list box will scroll to the next data set whose name begins with that letter. You can access additional **R** packages that are installed in your package library by *Tools* → *Load packages*.

4 Creating Numerical Summaries and Graphs

Once there is an active data set, you can use the **R Commander** menus to produce a variety of numerical summaries and graphs. I will describe just a few basic examples here. A good GUI should be largely self-explanatory. If you prefer a more extensive introduction to the **R Commander**, see Fox (2017).

In the initial examples below, I assume that the active data set is the `Nations` data set, read from a text file in the previous section. If you typed in the five-observation data set from Moore (2000), or read in the `Prestige` data set from the `car` package — operations that were also described in the previous section — then one of these is the active data set. Recall that you can change the active data set by clicking on the flat button with the active data set’s name near the top left of the *R Commander* window, choosing from among a list of data sets currently resident in memory.

Selecting *Statistics* → *Summaries* → *Active data set* produces the results shown in Figure 8. For each numerical variable in the data set (`TFR`, `contraception`, `infant.mortality`, and `GDP`), **R** reports the minimum and maximum values, the first and third quartiles, the median, and the mean, along with the number of missing values. For the categorical variable `region`, I get the number of observations at each “level” of the factor. Had the data set included more than ten variables, the **R Commander** would have asked me whether I really want to proceed — potentially protecting me from producing unwanted voluminous output. This menu item is unusual in that it directly invokes an **R** command rather than leading to a dialog box, as is more typical of **R Commander** menus.

For example, selecting *Statistics* → *Summaries* → *Numerical summaries...* brings up the dialog box in Figure 9. Only numerical variables appear in the variable list in this dialog; the factor `region` is excluded

¹⁴Not all data in packages are data frames but only data frames are suitable for use in the **R Commander**. If you attempt to read data that are not in a data frame, the **R Commander** will try to change them into a data frame, printing a warning message if it’s successful and an error message if it’s not.

¹⁵The example shows data sets, including `Prestige`, residing in the `car` package; these data sets currently reside in the `carData` package, which is loaded along with the `car` package when the **R Commander** starts up.

¹⁶In general in the **R Commander**, when it is necessary to copy an item from a list box to another location in a dialog, a double-click is required.

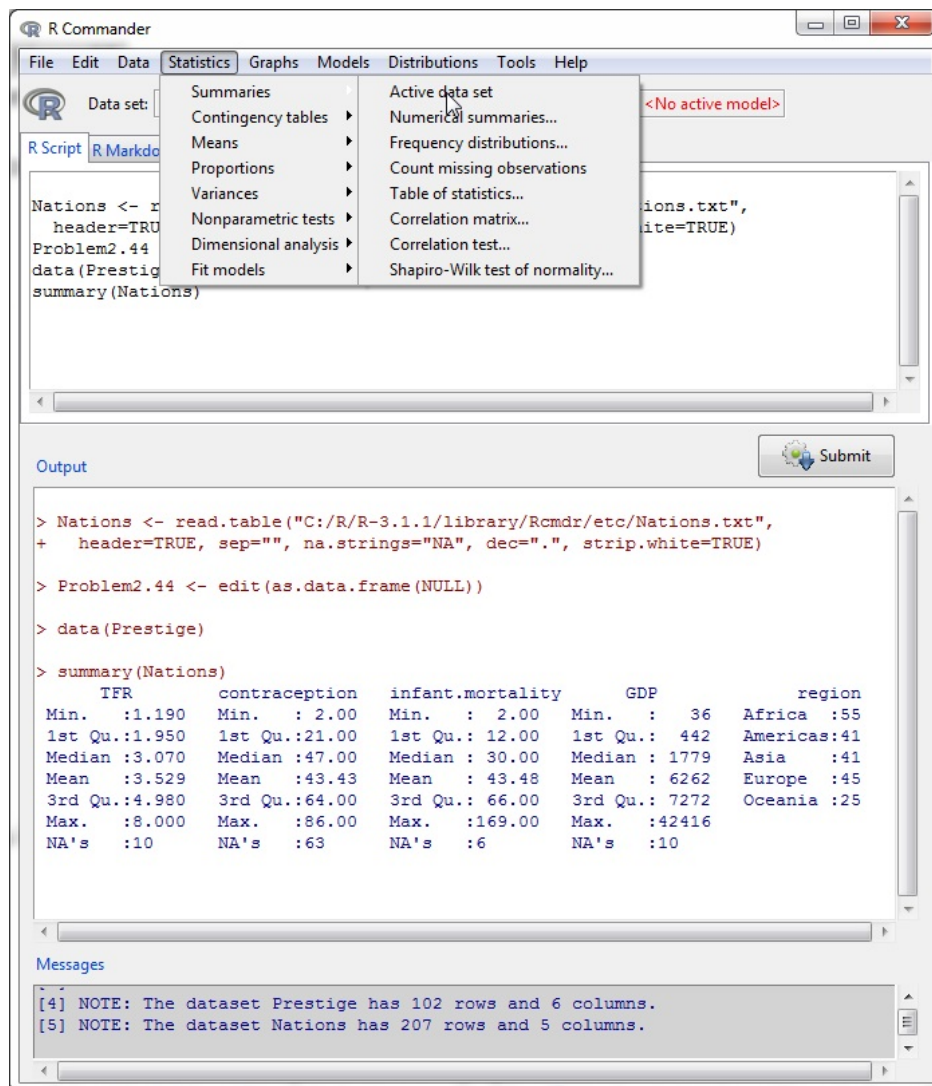


Figure 8: Getting variable summaries for the active data set.

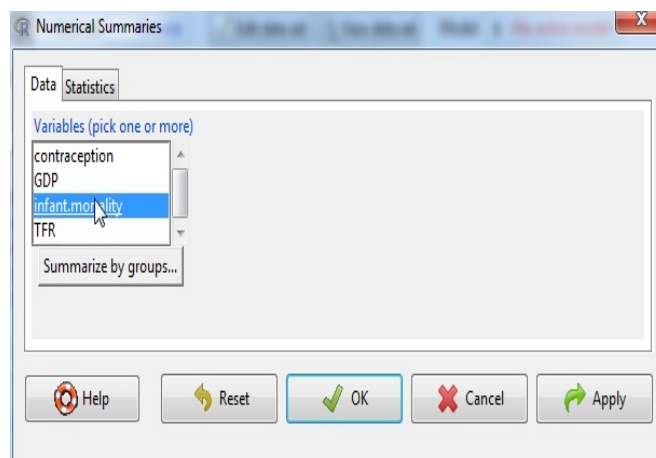


Figure 9: The *Data* tab in the *Numerical Summaries* dialog box.

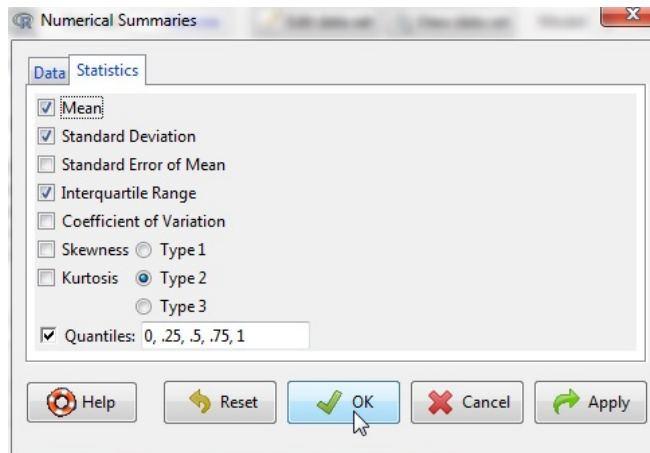


Figure 10: The *Statistics* tab in the *Numerical Summaries* dialog box.

because it is not sensible to compute numerical summaries like the mean and standard deviation for a factor. I selected the variable `infant.mortality` by left-clicking on it.¹⁷ The *Numerical Summaries* dialog box has two tabs: *Data* and *Statistics*. Click on the *Statistics* tab to select it, as shown in Figure 10. In this case, I'll take all of the default statistics selections. Clicking *OK*, produces the following output (in the *Output* pane):

```
> numSummary(Nations[, "infant.mortality"], statistics=c("mean", "sd", "IQR",
+ "quantiles"), quantiles=c(0, .25, .5, .75, 1))
      mean      sd IQR 0% 25% 50% 75% 100%    n NA
43.47761 38.75604  54  2  12  30  66  169 201  6
```

By default, the **R** command that is executed prints out the mean, standard deviation (*sd*), and interquartile range (*IQR*) of the variable, along with quantiles (percentiles) corresponding to the minimum, the first quartile, the median, the third quartile, and the maximum; *n* is the number of valid observations, and *NA* the number of missing values.

As is typical of **R Commander** dialogs, the *Numerical Summaries* dialog box in Figure 9 includes *Help*, *Reset*, *OK*, *Cancel*, and *Apply* buttons.¹⁸ The *Help* button leads to a help page (which appears in your web browser) either for the dialog itself or (as here) for an **R** function that the dialog invokes. The *Reset* button, which is present in most **R Commander** dialogs, resets the dialog to its original state; otherwise, the dialog retains selections from a previous invocation. Dialog state is also reset when the active data set changes. As demonstrated, the *OK* button closes the dialog and generates an **R** command. The *Apply* button also generates a command, but then reopens the dialog in its current state, facilitating the application of several similar operations. If you make an error in a dialog box — for example, clicking *OK* without choosing a variable in the *Numerical Summaries* dialog — an error message will typically appear and the dialog box will reopen.

The *Numerical Summaries* dialog box also makes provision for computing summaries within groups defined by the levels of a factor. Clicking on the *Summarize by groups...* button in the *Data* tab brings up the *Groups* dialog, as shown in Figure 11. Because there is only one factor in the *Nations* data set, only the variable `region` appears in the variable list, and it is pre-selected; clicking *OK* changes the *Summarize by groups...* button to *Summarize by region* (see Figure 12). In this case, I have selected two numerical variables to summarize, `GDP` and `infant.mortality`. Clicking *OK* produces the following results in the *Output* pane:

¹⁷To select a single variable in a variable-list box, simply left-click on its name. In some contexts, you will have to (or want to) select more than one variable. In these cases, the usual **Windows** conventions apply: Left-clicking on a variable selects it and de-selects any variables that have previously been selected; *Shift-left-click* extends the selection; and *Ctrl-left-click* toggles the selection for an individual variable.

¹⁸The order of the buttons varies according to the operating system, and is different, for example, in **macOS** than in **Windows**.

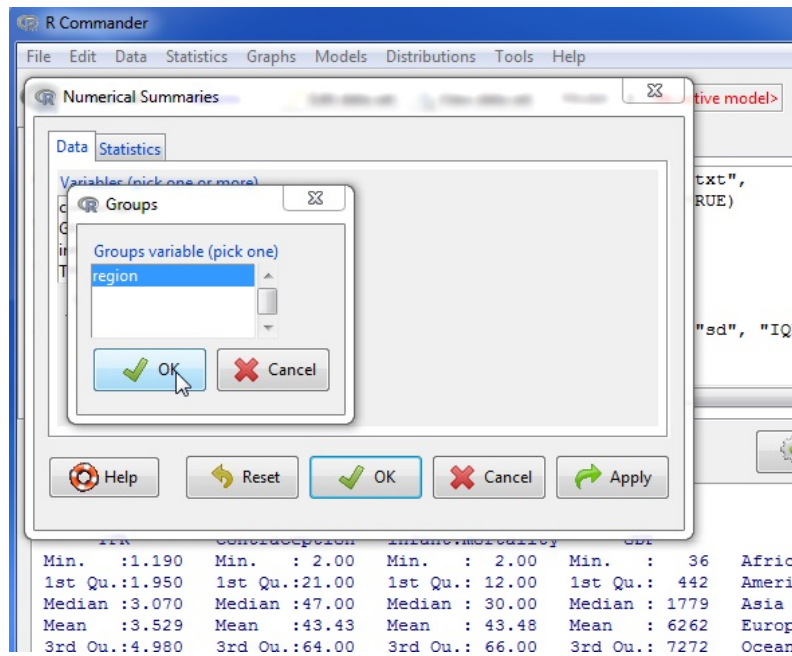


Figure 11: Selecting a grouping variable in the *Groups* dialog box.

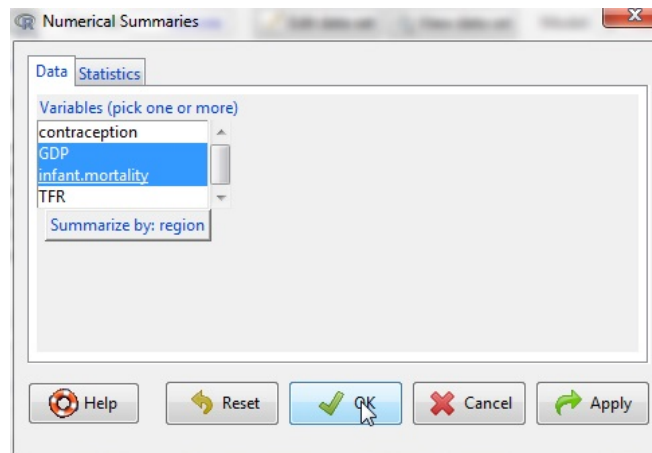


Figure 12: The *Numerical Summaries* dialog box after the grouping variable `region` has been chosen and with two numeric variables selected.

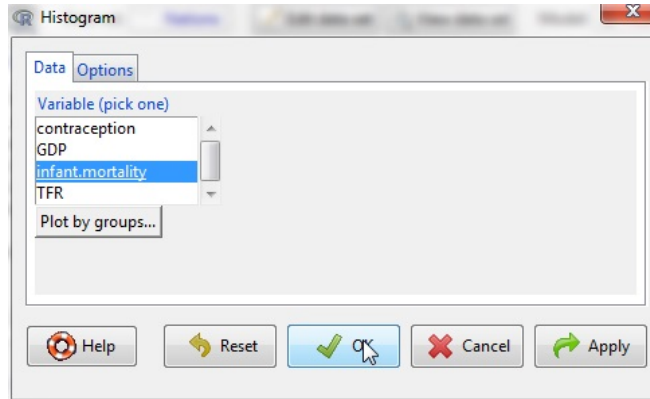


Figure 13: The *Histogram* dialog.

```
> numSummary(Nations[,c("GDP", "infant.mortality")], groups=Nations$region,
+   statistics=c("mean", "sd", "IQR", "quantiles"), quantiles=c(0,.25,.5,.75,1))
```

```
Variable: GDP
```

	mean	sd	IQR	0%	25%	50%	75%	100%	n	NA
Africa	1196.000	2089.614	795.50	36	209.00	389.5	1004.50	11854	54	1
Americas	5398.000	6083.311	5268.50	386	1749.25	2765.5	7017.75	26037	40	1
Asia	4505.051	6277.738	6062.50	122	345.00	1079.0	6407.50	22898	39	2
Europe	13698.909	13165.412	24582.25	271	1643.75	9222.5	26226.00	42416	44	1
Oceania	8732.600	11328.708	16409.25	654	1102.75	2348.5	17512.00	41718	20	5

```
Variable: infant.mortality
```

	mean	sd	IQR	0%	25%	50%	75%	100%	n	NA
Africa	85.27273	35.188095	50.0	7	61.00	85.0	111.00	169	55	0
Americas	25.60000	17.439713	24.0	6	12.00	21.5	36.00	82	40	1
Asia	45.65854	32.980001	50.0	5	22.00	37.0	72.00	154	41	0
Europe	11.85366	7.122363	10.0	5	6.00	8.0	16.00	32	41	4
Oceania	27.79167	29.622229	26.5	2	9.25	20.0	35.75	135	24	1

Several other **R Commander** dialogs allow you to select a grouping variable in this manner.

Making graphs with the **R Commander** is also straightforward. For example, selecting *Graphs* → *Histogram...* from the **R Commander** menus produces the *Histogram* dialog box in Figure 13. There are *Data* and *Options* tabs in this dialog. I'll take all the default options (the *Options* tab isn't shown), and clicking on *infant.mortality* followed by *OK*, opens a *Graphics Device* window with the histogram shown in Figure 14. If you make several graphs in a session, then only the most recent normally appears in the *Graphics Device* window.¹⁹

5 Statistical Models

Several kinds of statistical models can be fit in the **R Commander** using menu items under *Statistics* → *Fit models*: linear models (by both *Linear regression* and *Linear model*), generalized linear models, multinomial logit models, ordinal regression models such as the proportional-odds model [the latter two from Venables and Ripley's **nnet** and **MASS** packages, respectively (Venables and Ripley, 2002)], and linear and generalized linear models [using the **lme4** package (Bates et al., 2015)]. Although the resulting

¹⁹On **Windows**, you can recall previous graphs using the *Page Up* and *Page Down* keys on your keyboard if you first turn on the graph history feature of the **Windows R** graphics device, via *History* → *Recording*. In **macOS**, the left and right arrows may be used to scroll through graphs. Dynamic three-dimensional graphs, created, for example, by *Graphs* → *3D graph* → *3D scatterplot...* appear in a special *RGL device* window.

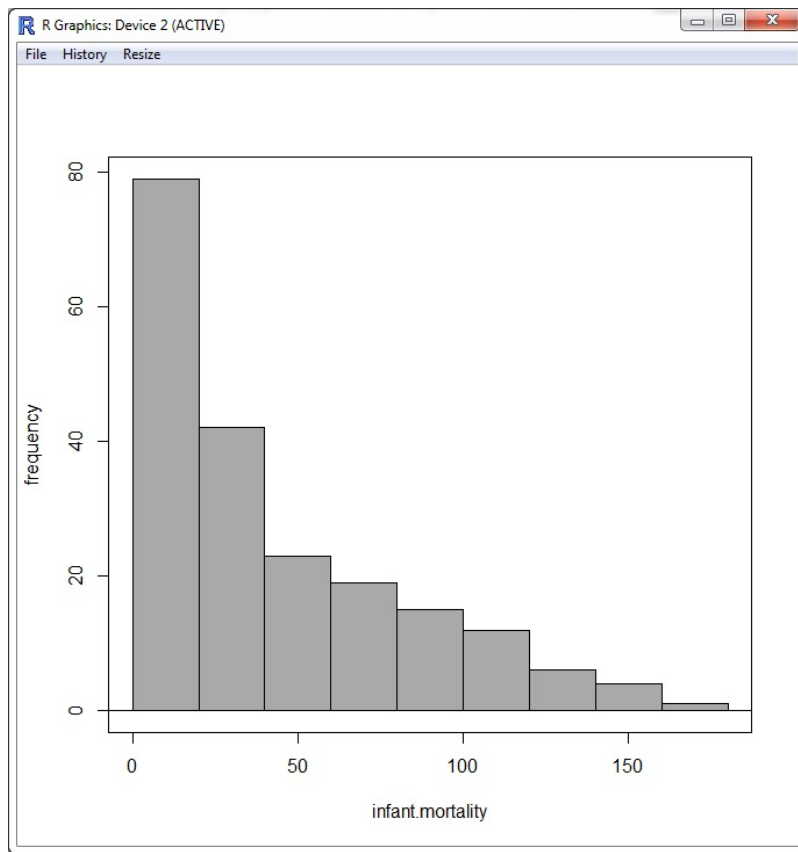


Figure 14: A graphics window containing the histogram for infant mortality in the **Nations** data set.

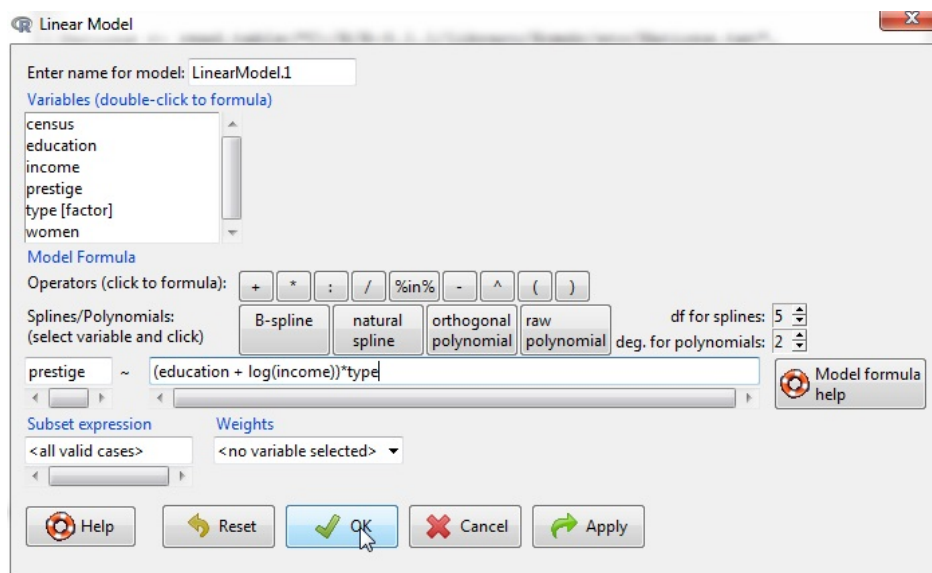


Figure 15: The *Linear Model* dialog box, with **Prestige** from the **car** package as the active data set.

dialog boxes differ in certain details (for example, the generalized linear model dialog makes provision for selecting a distributional family and corresponding link function), they share a common general structure, as illustrated in the *Linear Model* dialog in Figure 15.²⁰ Before selecting *Statistics* → *Fit models* → *Linear Model*, I made **Prestige** the active data set by clicking on the active data set button and selecting **Prestige** from the resulting list. Recall that the **Prestige** data were read from the **car** package in Section 3.3.

- Double-clicking on a variable in the variable-list box copies it to the model formula — to the left-hand side of the formula, if it is empty, otherwise to the right-hand side (with a preceding + sign if the context requires it). Factors (categorical variables — here, **type**) are parenthetically labelled as such in the variable list.²¹ Entering a factor into the right-hand side of a statistical model formula generates dummy-variable regressors.
- The top row of buttons in the toolbar above the formula can be used to enter operators and parentheses into the right-hand side of the formula.
- The bottom row of buttons in the toolbar can be conveniently used to enter regression-spline and polynomial terms into the model formula, with the degrees of freedom for splines and the degree of polynomials controlled by the spin-boxes to the right of the buttons (defaulting to 5 df and degree 2, respectively).
- You can also type directly into the formula fields, and indeed may have to do so, for example, to put a term such as `log(income)` into the formula, as I've done here. Some information on **R** model formulas may be obtained by pressing the *Model formula help* button in the linear-model dialog.
- The name of the model, here **LinearModel.1**, is automatically generated, but you can substitute any valid **R** name.
- You can type an **R** expression into the box labelled *Subset expression*; if supplied, this is passed to the **subset** argument of the **lm** function, and is used to fit the model to a subset of the observations in the data set. One form of subset expression is a logical expression that evaluates to **TRUE** or **FALSE** for

²⁰An exception is the *Linear Regression* dialog in which the response variable and explanatory variables are simply selected by name from list boxes containing the numeric variables in the current data set.

²¹Some data frames contain logical variables (with values **TRUE** and **FALSE**) and character variables, with values that are text strings (such as "male" and "female"). If such variables are present, the R Commander will treat them as if they were factors. In most context, this will work properly. Character data read from plain-text files will automatically be converted to factors.

each observation, such as `type != "prof"` (which would select all non-professional occupations from the `Prestige` data set).

- In version 2.8-0 of the **R Commander** (or later), there's also a text box into which you can type the names or numbers of cases (rows) to be omitted from the fitted model. (This box isn't shown in Figure 15.)
- Optionally selecting a weight variable in the *Weights* drop-down list produces a weighted-least-squares (WLS) regression.

Clicking the *OK* button generates the following command and output, and makes `LinearModel.1` the active model, with its name displayed in the *Model* button:

```
> LinearModel.1 <- lm(prestige ~ (education + log(income))*type,
+   data=Prestige)

> summary(LinearModel.1)

Call:
lm(formula = prestige ~ (education + log(income)) * type, data = Prestige)

Residuals:
    Min       1Q   Median       3Q      Max
-13.970  -4.124   1.206   3.829  18.059

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   -120.0459    20.1576  -5.955 5.07e-08 ***
education         2.3357     0.9277   2.518 0.01360 *
log(income)     15.9825     2.6059   6.133 2.32e-08 ***
type[T.prof]     85.1601    31.1810   2.731 0.00761 **
type[T.wc]       30.2412    37.9788   0.796 0.42800
education:type[T.prof]  0.6974     1.2895   0.541 0.58998
education:type[T.wc]   3.6400     1.7589   2.069 0.04140 *
log(income):type[T.prof] -9.4288     3.7751  -2.498 0.01434 *
log(income):type[T.wc] -8.1556     4.4029  -1.852 0.06730 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.409 on 89 degrees of freedom
(4 observations deleted due to missingness)
Multiple R-squared:  0.871, Adjusted R-squared:  0.8595
F-statistic: 75.15 on 8 and 89 DF,  p-value: < 2.2e-16
```

Operations on the active model may be selected from the *Models* menu. For example, *Models* → *Hypothesis tests* → *Anova table...*, followed by selecting the default “Type-II” tests, produces the following output:

```
> Anova(LinearModel.1, type="II")
Anova Table (Type II tests)

Response: prestige
              Sum Sq Df F value    Pr(>F)
education    1209.3  1 29.4446 4.912e-07 ***
log(income)  1690.8  1 41.1670 6.589e-09 ***
type         469.1  2  5.7103 0.004642 **
```

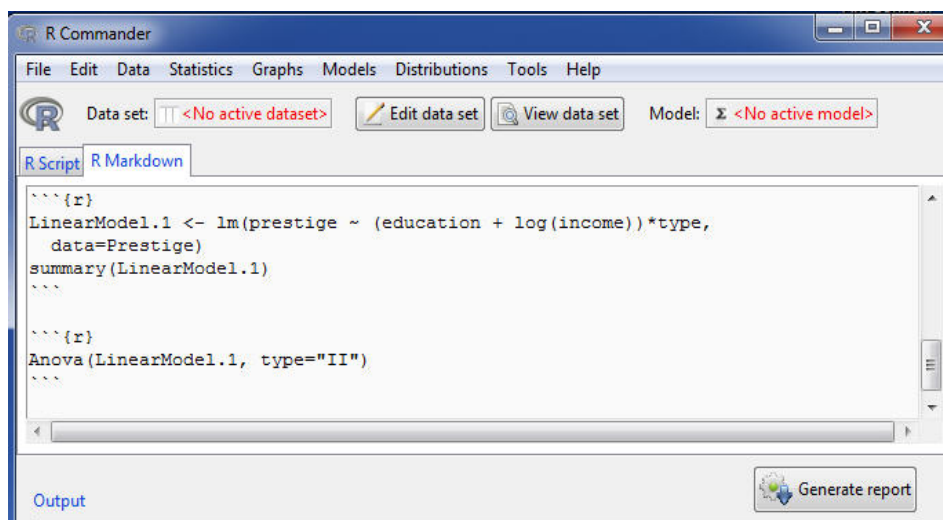


Figure 16: The *R Markdown* tab, with *Generate report* button.

```
education:type      178.8  2  2.1762  0.119474
log(income):type    290.3  2  3.5344  0.033338 *
Residuals          3655.4 89
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6 Odds and Ends

6.1 Producing Reports

In its default configuration, the **R Commander** includes an *R Markdown* tab in the upper pane, which accumulates the commands generated during the session in an **R Markdown** document.²² Figure 16 shows the *R Markdown* tab for the current session, which I’ve scrolled to the top of the generated document. As its name implies, **R Markdown** is a simple markup language, which includes blocks of **R** commands, and can be used to generate HTML (i.e., web) pages or other types of documents. For more information about **R Markdown**, select *Help* → *Using R Markdown* from the **R Commander** menus.²³

Each set of commands generated by the **R Commander** produces a block of **R** commands in the **R Markdown** document.²⁴ These blocks are delimited by ````\{r}` at the start of each block and by ````` (three back-ticks) at the end of the block. Even if you are unfamiliar with **R** commands, you can see the relationship between commands and resulting output in the *Output* pane.

The *R Markdown* tab is editable, and so you can modify and add to the text in the tab. It is generally safe to type whatever explanatory text you wish *between* blocks of **R** code (see below). In general, however, unless you know what you’re doing, you should not modify **R** code blocks in the **R Markdown** document or add your own code blocks. You can, however, remove entire blocks of code, as long as subsequent blocks don’t depend on them — and you will likely want to remove command blocks that produce unwanted output. Command blocks that produce errors are removed automatically. You can remove the most recent command block by selecting *Remove last Markdown Command Block* from the *Edit* menu, or by right-clicking in the *R Markdown* tab and selecting *Remove last Markdown Command Block* from the context menu. The first

²²The **R Commander** can also optionally create a **knitr LaTeX** document (Xie, 2016) and compile it into a PDF file. This option requires a **LaTeX** installation, and is activated by setting the **Rcmdr** option `use.knitr` to `TRUE`. see *Help* → *Commander help* and *Tools* → *Options*.

²³This and some other items in the *Help* menu require an active internet connection.

²⁴Commands that require direct user interaction, such as interactive point identification in a graph, are suppressed in the **R Markdown** document. As well, commands that generate errors are removed from the document.

code block (which begins ````{r echo=FALSE}`) sets some options for the software from the **knitr** package (Xie, 2013) that is used to process the **R Markdown** text, and this block should not normally be modified.

With some lines elided (indicated by `. . .`), here is the **R Markdown** document produced for the current session:

```
---
title: "Replace with Main Title"
author: "John Fox"
date: "`r Sys.Date()`" # Uses current date
---

```{r echo=FALSE, message=FALSE}
include this code chunk as-is to set options
knitr::opts_chunk$set(comment=NA, prompt=TRUE)
library(Rcmdr)
library(car)
library(RcmdrMisc)
```

```{r echo=FALSE}
include this code chunk as-is to enable 3D graphs
library(rgl)
knitr::knit_hooks$set(webgl = hook_webgl)
```

```{r}
Nations <- read.table("C:/R/R-3.3.2/library/Rcmdr/etc/Nations.txt",
 header=TRUE, sep="", na.strings="NA", dec=".", strip.white=TRUE)
```

. . .

```{r}
data(Prestige, package="car")
```

. . .

Let us regress occupational prestige on the education and income levels of the occupations,
transforming income to linearize its relationship to prestige:

### Linear Model: LinearModel.1: prestige ~ (education + log(income))*type
```{r}
LinearModel.1 <- lm(prestige ~ (education + log(income))*type,
 data=Prestige)
summary(LinearModel.1)
```

### Analysis of Variance: LinearModel.1
```{r}
Anova(LinearModel.1, type="II")
```

...

It is probably unnecessary to explain that you would normally replace “Replace with Main Title” with the title of the report that you want to create. Perhaps less obviously, you can type arbitrary explanatory text before the first command block, *in between* **R** code blocks — that is, between the terminating ````` of one block and starting ````{r}` of the next — and after the last command block. You can take advantage of the simple markup provided by **R Markdown**; for example, text enclosed in asterisks (e.g., *\*this is important\**) will be set in *italic* type. To illustrate, I added the text “Let us regress occupational prestige ...” immediately before the **R** command block performing the regression.

The **R Markdown** document also includes automatically generated section headings for some **R** command blocks. These are preceded by three hashmarks (###), which indicate a level-3 heading. When the **R Markdown** document is converted into a report (see immediately below), the headers appear not only in the text but in an automatically generated index. You can edit these headings to customize them, and you can also add your own headings, using, for example, # for a level-1 heading and ## for a level-2 heading.

Once you have finished editing the **R Markdown** document, you can generate a report from it in the form of an HTML document (web page), **Word** document, rich text file, or PDF file by pressing the *Generate report* button below the *R Markdown* tab.<sup>25</sup> An HTML report should open in your web browser, and a PDF report in your PDF viewer. **Word** and rich text files must be opened manually, but can easily be edited further. The **R Markdown** document can be saved via the *File* menu.

You can also, and more conveniently, open a separate and larger editor window to edit the **R Markdown** document (see Figure 17): Select *Edit R Markdown document* from the **R Commander Edit** menu; right-click in the *R Markdown* tab and select *Edit R Markdown document* from the context menu; or press the key-combination *Control-E* when the cursor is in the *R Markdown* tab. The editor supports the usual *Edit*-menu and right-click context-menu commands, and also allows you to compile the **R Markdown** document into an HTML report. Clicking the *OK* button in the editor saves your edits to the *R Markdown* tab, and clicking *Cancel* discards your edits. Below the menu in the editor, there is a largely self-explanatory toolbar with various buttons; if you hover the mouse over a button, a “tool-tip” will display.

You can open the **R Markdown** editor window at the beginning of a session and leave it open while you work. Commands generated by the **R Commander** will be entered both into the *R Markdown* tab and the editor, and you’ll be able to write text in the editor as you go. See the editor help for more information.

## 6.2 Saving and Printing Output

You can also save text output directly from the *File* menu in the **R Commander**; likewise you can save or print a graph from the *File* menu in an **R Graphics Device** window. If you prefer not to use the *R Markdown* tab, you can collect the text output and graphs that you want to keep in a word-processor document. In this manner, you can intersperse **R** output with your typed notes and explanations. This procedure, however, has the disadvantage that it is not directly reproducible, while an **R Markdown** document can subsequently be executed to reproduce your analysis, possibly with modifications.

Open a word processor such as **Word**, **OpenOffice Writer**, or even **Windows WordPad**. To copy text from the *Output* pane, block the text with the mouse, select *Copy* from the *Edit* menu (or press the key combination *Ctrl-c*, or right-click in the pane and select *Copy* from the context menu), and then paste the text into the word-processor document via *Edit* → *Paste* (or *Ctrl-v* or right-click *Paste*), as you would for any **Windows** application. One point worth mentioning is that you should use a monospaced (“**typewriter**”) font, such as *Courier New*, for text output from **R**; otherwise the output will not line up neatly.

Likewise, to copy a graph, select *File* → *Copy to the clipboard* → *as a Metafile* from the **R Graphics Device** menus; then paste the graph into the word-processor document via *Edit* → *Paste* (or *Ctrl-v* or right-click *Paste*). Alternatively, you can use *Ctrl-w* to copy the graph from the **R Graphics Device**, or right-click on the graph to bring up a context menu, from which you can select *Copy as metafile*.<sup>26</sup> At the

<sup>25</sup>To generate **Word** or rich text file documents you must install **Pandoc**; to generate PDF files, you’ll additionally need **L<sup>A</sup>T<sub>E</sub>X**. To install this optional auxiliary software, use the *Tools* menu.

<sup>26</sup>As you will see when you examine these menus, you can save graphs in a variety of formats, and to files as well as to the clipboard. The procedure suggested here is straightforward, however, and generally results in high-quality graphs. Once again, this description applies to **Windows** systems.

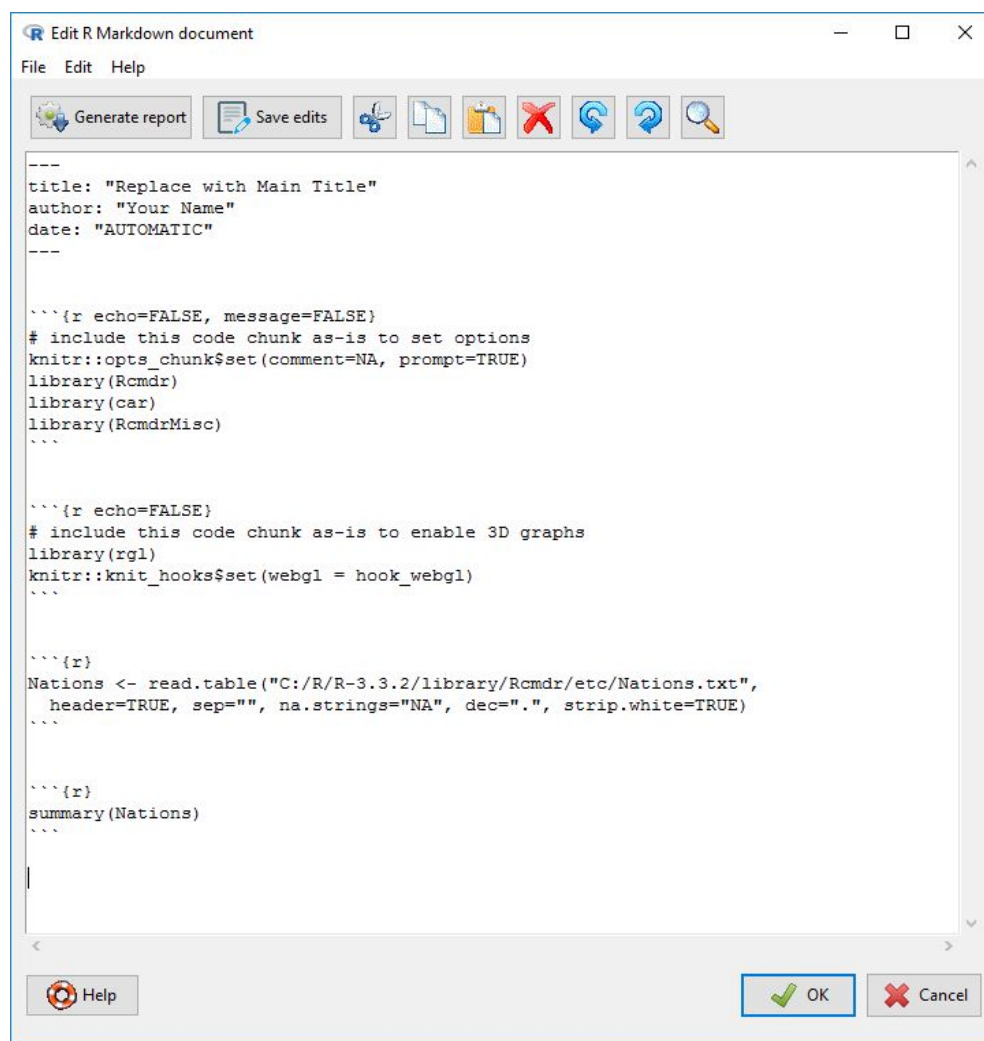


Figure 17: The **R Markdown** document editor. (Note: This image of the editor was created in an earlier version of the **R Commander**, which didn't automatically generate level-3 section headings for commands.)

end of your **R** session, you can save or print the document that you have created, providing an annotated record of your work.

Alternative routes to saving text and graphical output may be found respectively under the **R Commander** *File* and *Graphs* → *Save graph to file* menus. Saving the **R Commander** *Script* tab, via *File* → *Save script*, allows you to reproduce your work on a future occasion.

### 6.3 Entering Commands in the Script Tab

The *R Script* tab provides a simple facility for editing, entering, and executing commands. Commands generated by the **R Commander** appear in the *Script* tab, and you can type and edit commands in the tab more or less as in any editor. The **R Commander** does not provide a true “console” for **R**, however, and the *Script* tab has some limitations. For example, all lines of a multiline command must be submitted simultaneously for execution. For serious **R** programming, it is preferable to use the script editors provided by the **Windows** and **macOS** versions of **R**, or — even better — a programming editor or interactive development environment, such as **RStudio** <posit.co/>.<sup>27</sup>

### 6.4 Using R Commander Plug-ins

**R Commander** plug-ins are **R** packages that add to the capabilities of the **R Commander**. Many such plug-ins are currently available on CRAN, and may be downloaded and installed in the normal manner. Plug-ins typically add menus or menu items and associated dialog boxes to the **R Commander**. They may also modify or remove existing menu items or dialogs. A properly programmed **R Commander** plug-in can be loaded either directly, in which case the **R Commander** loads along with the plug-in, or from the **R Commander** via *Tools* → *Load Rcmdr plug-in(s)*... . In the latter event, the **R Commander** will restart to activate the plug-in. It is possible to use several plug-ins simultaneously, but plug-ins may also conflict with each other. For example, one plug-in may remove a menu to which another plug-in tries to add a menu item.

### 6.5 Terminating the R Session

There are several ways to terminate your session. For example, you can select *File* → *Exit* → *From Commander and R* from the **R Commander** menus. You will be asked to confirm, and then asked whether you want to save the contents of the *R Script*, *Output*, and *R Markdown* windows. Likewise, you can select *File* → *Exit* from the *R Console*; in this case, you will be asked whether you want to save the **R** workspace (i.e., the data that **R** keeps in memory); you would normally answer *No*: Saving the **R** workspace will likely create problems in subsequent sessions.

## References

- Bates, D., Mächler, M., Bolker, B., and Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1):1–48.
- Firth, D. (2016). *relimp: Relative Contribution of Effects in a Regression Model*. R package version 1.0-5.
- Fox, J. (2005). The R Commander: A basic-statistics graphical user interface to R. *Journal of Statistical Software*, 19(9):1–42.
- Fox, J. (2007). Extending the R Commander by “plug-in” packages. *R News*, 7(3):46–52.
- Fox, J. (2017). *Using the R Commander: A Point-and-Click Interface for R*. Chapman & Hall/CRC Press, Boca Raton FL.
- Fox, J. and Sá Carvalho, M. (2012). The RcmdrPlugin.survival package: Extending the R Commander to survival analysis. *Journal of Statistical Software*, 49(7):1–32.

---

<sup>27</sup>The **R Commander** will run under **RStudio**, in which case by default **R Commander** output and messages are directed to the **R** console within **RStudio**.

- Moore, D. S. (2000). *The Basic Practice of Statistics*. Freeman, New York, second edition.
- R Core Team (2020). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Venables, W. N. and Ripley, B. D. (2002). *Modern Applied Statistics with S*. Springer, New York, fourth edition. ISBN 0-387-95457-0.
- Xie, Y. (2016). *knitr: A general-purpose package for dynamic report generation in R*. R package version 1.13.